

|                                                    |                                                           |
|----------------------------------------------------|-----------------------------------------------------------|
| Program: <b>Diploma in Automation and Robotics</b> |                                                           |
| Course Code:                                       | Course Title: <b>Electrical Circuits and Machines Lab</b> |
| Semester: <b>3</b>                                 | Credits: <b>2.5</b>                                       |
| Course Category: <b>Program Core</b>               |                                                           |
| Periods per week: <b>4 (L:1, T:0, P:3)</b>         | Periods per semester: <b>60</b>                           |

### Course Objectives:

- Provide working knowledge for the analysis of basic DC and AC circuits
- Provide the practical exposure regarding construction and operation of various electrical machines like DC generators, DC Motors, Induction Motors and Transformers.

### Course Prerequisites:

| Topic                                                                                                                                                                                                     | Course Code | Course Name                                                | Semester |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------|----------|
| Basic principles and theorems of Engineering Mathematics                                                                                                                                                  |             | Mathematics I &II                                          | 1 & 2    |
| Skills in measurements and testing of electric circuits using different meters like Multimeter, DC voltmeter, ammeter and wattmeter                                                                       | 2039        | Fundamentals of Electrical and Electronics Engineering Lab | 2        |
| Fundamental concepts and theorems for the analysis of basic AC and DC circuits, Construction and operation of various electrical machines like DC Generator, DC Motors, Induction Motors and Transformers |             | Electrical Circuits and Machines                           | 3        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                               | Duration (Hours) | Cognitive Level |
|-----------------|-------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Apply fundamental concepts and circuit laws to solve simple DC electric circuits          | 15               | Applying        |
| CO2             | Realize the concept of resonance and learn the power measurement by two wattmeter method. | 12               | Applying        |

|     |                                                                |    |          |
|-----|----------------------------------------------------------------|----|----------|
| CO3 | Learn the construction and operation of DC generator and motor | 15 | Applying |
| CO4 | Learn the working and testing methods of transformer           | 12 | Applying |
|     | Lab Test                                                       | 6  |          |

### CO-PO Mapping:

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             | 3   |     |     |     |     |     |     |
| CO3             | 3   |     |     |     |     |     |     |
| CO4             | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

### Course Outline:

| Module Outcomes | Description                                                                                        | Duration (Hours) | Cognitive Level |
|-----------------|----------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | <b>Apply fundamental concepts and circuit laws to solve simple DC electric circuits</b>            |                  |                 |
| M1.01           | Verify Kirchhoff 's current Law and Kirchhoff 's Voltage Law                                       | 3                | Applying        |
| M1.03           | Test and Verify Thevenin's and Norton theorem                                                      | 3                | Applying        |
| M1.05           | Test and verify the super position theorem                                                         | 3                | Applying        |
| M1.06           | Measure the power in RLC parallel circuit. Calculate the power factor and draw the phasor diagram. | 3                | Applying        |
| M1.07           | Measure the power in RLC series circuit. Calculate the power factor and draw the phasor diagram.   | 3                | Applying        |
| CO2             | <b>Realize the concept of resonance and power measurement by two wattmeter method.</b>             |                  |                 |
| M2.01           | Construct RLC Circuit for Series Resonance and draw the Frequency versus Impedance curve           | 3                | Applying        |
| M2.02           | Construct RLC Circuit for Parallel Resonance and draw the Frequency versus Impedance curve         | 3                | Applying        |

|            |                                                                                                            |   |          |
|------------|------------------------------------------------------------------------------------------------------------|---|----------|
| M2.03      | Measurement of 3 $\phi$ power by two wattmeter method                                                      | 6 | Applying |
|            | Lab Test 1                                                                                                 | 3 |          |
| <b>CO3</b> | <b>Learn the working and speed control methods of DC generator and motor.</b>                              |   |          |
| M3.01      | Determination of Open Circuit Characteristics of DC shunt generator                                        | 6 | Applying |
| M3.02      | Conduct Load test on DC shunt generator and draw the performance curve                                     | 6 | Applying |
| M3.03      | Conduct the load test on DC series motor and draw the performance curve                                    | 3 | Applying |
| <b>CO4</b> | <b>Learn the working and testing methods of transformer and working of induction motor</b>                 |   |          |
| M4.01      | Conduct the Load test on 1 $\phi$ Transformer                                                              | 3 | Applying |
| M4.02      | Predetermine the efficiency and regulation by open circuit and short circuit tests on 1 $\phi$ transformer | 3 | Applying |
| M4.03      | Perform load test on 1 $\phi$ induction motor                                                              | 3 | Applying |
| M4.04      | Perform load test on 3 $\phi$ squirrel cage induction motor                                                | 3 | Applying |
|            | Lab Test 2                                                                                                 | 3 |          |

**Text / Reference:**

| <b>T/R</b> | <b>Book Title / Author</b>                                                       |
|------------|----------------------------------------------------------------------------------|
| T1         | Bhattacharya, S. K., Basic Electrical & Electronics Engineering, Pearson         |
| R1         | Nagoor kani , Circuit Theory ,RBA Publications                                   |
| R2         | Theraja. B.L., A text book of Electrical Technology, Vol. I & III, S.Chand & Co. |

**Online Resources:**

| <b>Sl No</b> | <b>Website Link</b>                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://www.electrical4u.com/">https://www.electrical4u.com/</a>                                   |
| 2            | <a href="https://nptel.ac.in/courses/108/108/108108076/">https://nptel.ac.in/courses/108/108/108108076/</a> |